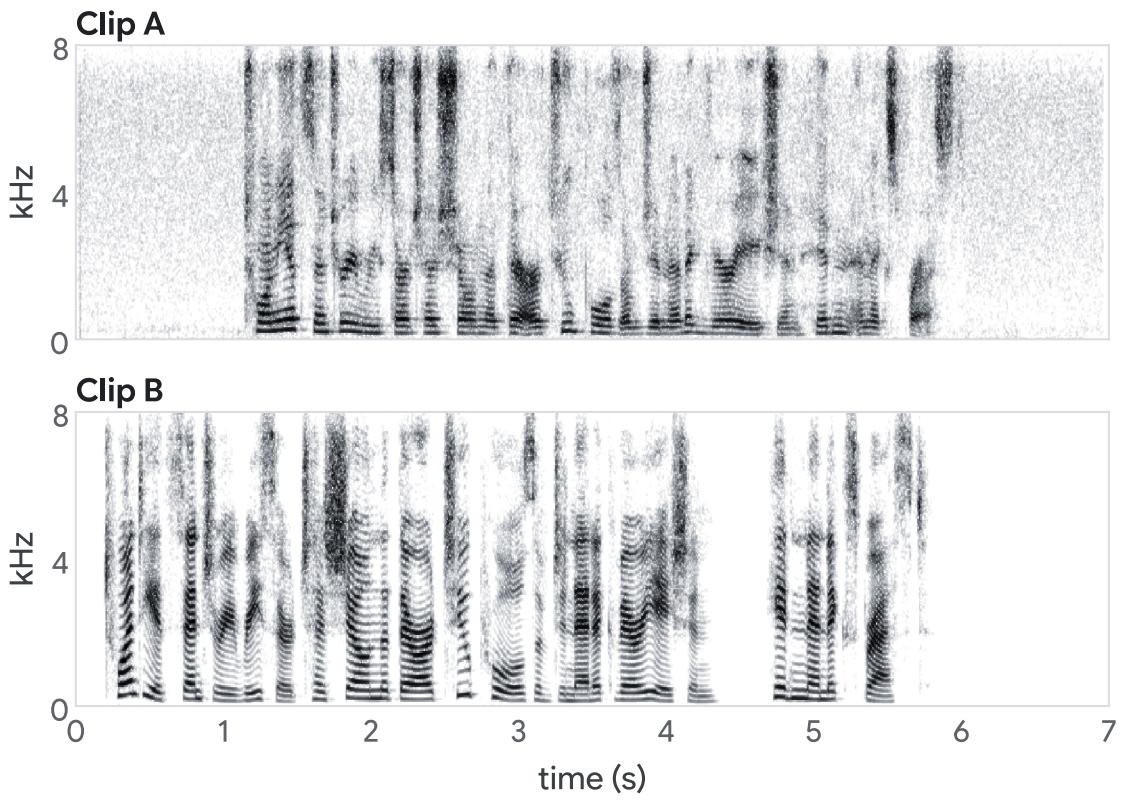
Audio Deepfake Detection

Is that voice note real?

Abdulrahman Tamim
Technical Director, GDG On Campus Abu Dhabi University
Sunday 11 October 2026, 4:30–6:00 PM, Microsoft Teams

A voice note arrives with no original. In 90 minutes you learn to read sound as numbers, see how a body makes a voice and how a machine imitates one, then build a detector in Google Colab, test it honestly and find where it fails.



Clips A and B say the same sentence. One is a recording of a person; the other was generated by software.

Before we start

Open the workshop notebook now. Its first cell checks that audio decoding and plotting work on your Colab runtime, so problems surface before we need them.

Four steps

1. Open the Colab link pinned in the Teams chat.
2. Sign in with a **personal** Google account; university accounts may block Colab.
3. Choose **File → Save a copy in Drive**, so your edits are kept.
4. Run the first cell, about 20 seconds. It plays a 4-second hiss and draws its spectrogram.

You need

A laptop with Chrome, a phone with WhatsApp, and headphones. Keep the Teams microphone muted; the notebook never uses your microphone.

What the first cell plays



To the ear it is noise. Its spectrogram spells a word written into the frequencies. Type the word in the chat: if you can read it, your setup works.

If something fails

A cell shows an error	Runtime → Restart session and run all
Upload does nothing	Drag the file into the Files panel on the left
Colab will not open	Use Chrome, or the read-only copy linked in the chat
Still stuck	Type “help” in the chat; a helper will message you

Point zero

Imagine your first week on a bank's fraud team. A manager forwards a WhatsApp voice note from "the CEO" asking for an urgent transfer, and asks you one question.

What you are given

file	AUD-20261011-WA0007.opus
duration	9.4 s
route	forwarded twice
original	not available
sender's phone	not available

What you are asked: "Is it really him? How sure are you? Can we act on it?" Each answer has a cost: a false alarm insults a real CEO, a miss sends the money.

What you need to know to answer

1. **Where the file has been.** Every app and speaker on the way changed it, and some evidence is gone before you start.
2. **How a body makes a voice.** Lungs, vocal folds, tongue and lips leave traces a generator has to imitate.
3. **How a machine makes one.** Each stage of speech synthesis can leave its own trace.
4. **How detectors work, and how often they are wrong.** We build one in Colab and measure it.
5. **What to do when detection is not enough.** Verification beats any score.

Scenario based on the 2020 UAE case on the next slide; file details are illustrative

The investigation

We follow the CEO voice note through six steps. Each step answers one question about it, and each one needs what the step before it taught us.

1 The file

Question: where has it been?

How calls, rooms and apps change a clip, and what they remove.

→

2 Sound

Question: what is a recording?

Sound as numbers, and the spectrogram: a picture of sound.

Colab Stop 1

→

3 Voice

Question: what does a body leave in a voice?

Vocal folds, the mouth as a filter, breathing.

→

4 Machine

Question: how is a voice cloned?

Three networks in a row, and the traces each leaves.

→

5 Detector

Question: can a program tell them apart, and when does it fail?

Measurements, a model, fair tests, training on the road.

Colab Stops 2 and 3

→

6 Verdict

Question: what do we tell the manager?

Base rates, a graded report, the call back.

Colab Stop 4

Each step builds on the last: a voice cannot be measured before we can read sound, and a detector cannot be trusted before we know what the road did to the file.

The Colab stops are where you check each step yourself, first on clips A and B, a person and a machine saying the same sentence, and at the end on your own voice note.

Voice deepfake cases

Four documented cases. In each one the voice was convincing enough to act on, and what gave it away was evidence outside the audio itself.

Case	How the voice was used	Channel	How it was caught	Outcome
UK energy company March 2019	Calls imitating the parent company's chief executive, German accent included, demanding an urgent payment.	phone calls	A third call came from an Austrian number; the manager grew suspicious.	€220,000 sent to a "Hungarian supplier".
UAE branch of a Japanese company Early 2020	A director's voice the manager knew, backed by emails from a "lawyer".	phone, email	Dubai Public Prosecution investigation; at least 17 people involved.	Up to $35 million transferred abroad.
Pikesville High School, USA January 2024	A fake recording of the principal making offensive remarks.	social media	Analysts said "probably AI"; account records identified the maker.	Athletic director: Alford plea, 4 months (2025).
New Hampshire primary, USA January 2024	A Biden-like voice told voters not to vote; it cost about $150 to make.	robocall	Pindrop traced it to an ElevenLabs-type voice; the consultant admitted it.	$6 million FCC fine; acquitted of criminal charges (2025).

Dubai Police, 7 October 2026: "*Stop. Verify. Then act.*"

GDRFA Dubai, 22 September 2026: "*visual or acoustic familiarity must no longer serve as proof of authenticity.*"

Gizmodo/WSJ 2019 · Forbes, 14 Oct 2021 · CBS Baltimore, Apr 2025 · Pindrop 2024; Bloomberg, Sep 2024; AP, Jun 2025 · Emirates 24/7, 7 Oct 2026 · UAE Vartha, 22 Sep 2026

Listening test

Both clips say the same 16-word sentence. One is a person reading aloud, the other is synthetic speech. Decide by ear before we measure anything.

A



6.97 s

B



6.65 s

The sentence: “Twentieth century research has shown that there are two pools of genetic variation: hidden and expressed.”

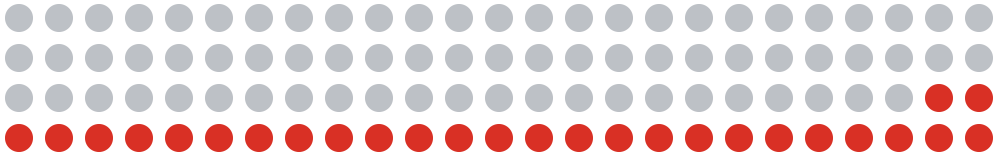
Vote in the Teams chat

Write A or B, your confidence from 1 (guess) to 5 (certain), and one thing you heard that decided it. Use headphones: laptop speakers hide breaths and hiss.

What to listen for

- **Breathing** between phrases, and where it falls
- **Timing:** pauses at meaning boundaries, or in odd places
- **Pitch:** rises and falls, or an even, flat melody
- **Background:** room sound, hiss, or total silence
- **Mouth sounds:** lip and tongue clicks, hesitations

How reliable are human ears?



Across 529 listeners, people identified 73 of every 100 deepfake clips; **27 passed as human**. Hearing example fakes first barely helped.

What listeners rely on

Mai and colleagues asked 529 people to label clips as real or fake and to explain why. The reasons people gave were the same whether their answer was right or wrong.

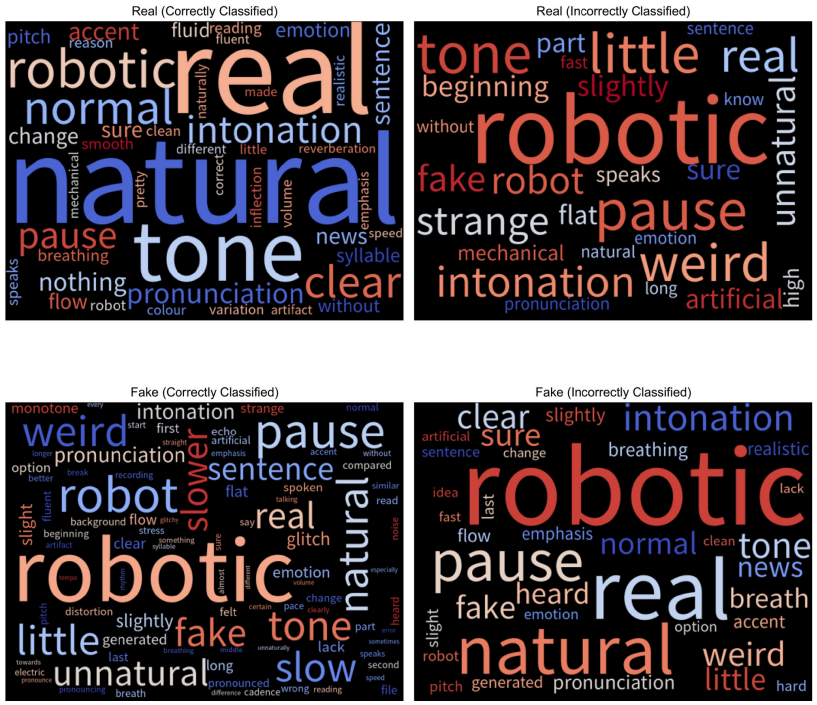


Fig. 9 from Mai et al., PLOS ONE 2023 (CC BY 4.0): reasons given by English-speaking participants. Top: real clips, bottom: fake clips; left: answered correctly, right: answered wrongly.

The result. Listeners caught 73% of deepfakes. Showing them examples of deepfakes first improved this only slightly.

Why ears fail. “Robotic”, “pause”, “tone” and “natural” appear in all four clouds, right or wrong. Modern systems pause naturally; real people sound flat on a phone.

Our clips. A is a person reading, from Google FLEURS. B was generated by Microsoft Edge text-to-speech. Compare with the chat vote.

The lesson. A feeling of “robotic” is not evidence. We need measurements we can explain to someone else.

The road a clip travels

Before anyone analyses a clip, it has usually passed through several machines. Each one keeps what people can hear and throws away detail a detector might have needed.

1 Source	→ 2 Air or line	→ 3 App	→ 4 Re-sharing	→ 5 You
A person speaks into a phone microphone. **A generator** writes a clean WAV, often 24 kHz.	**Played aloud** in a room and re-recorded. **A phone call,** compressed by a codec (AMR, G.711).	**WhatsApp note:** Opus at 16 kHz. **TikTok, Reels:** AAC with music and effects.	**Screen-recorded**, re-uploaded, forwarded again and again.	**Point zero:** one file, no original, no sender's device, no history.
The source: the only moment all the evidence exists.	Echo fills the pauses, noise hides breaths; a call keeps up to 3.4 kHz, 7 kHz on HD calls.	Nothing above 8 kHz; metadata stripped; loudness, speed and pitch changed.	Each step is one more lossy encoding and overwrites the encoder history.	Whatever the road removed cannot be recovered by any tool.

What usually remains: the words and the language, the melody (pitch), timing and pauses, formant movement below 4 kHz, the story the caller tells.
What usually does not: the original file, its metadata, fine detail above 8 kHz, true silence, the sender's device.
Codec: software that compresses audio for sending, such as Opus (WhatsApp), AAC (TikTok), AMR and G.711 (calls). **WAV:** uncompressed audio.

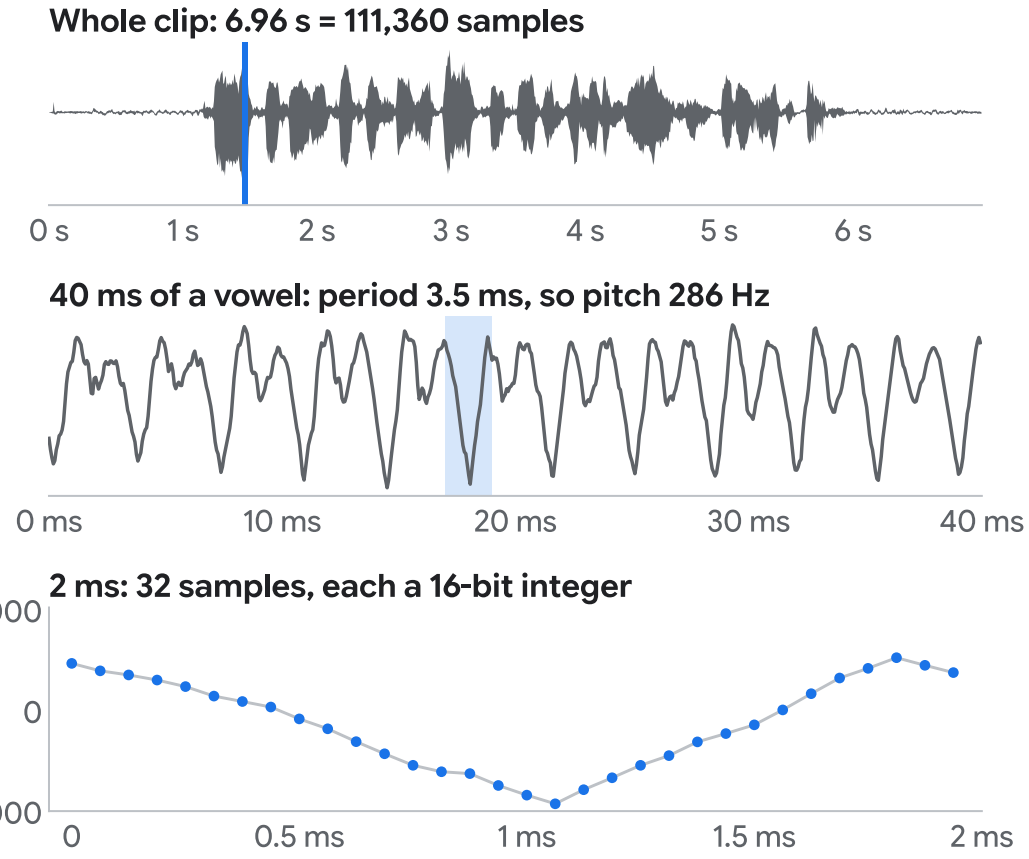
The first question is never "real or fake?" but "what happened to this file?". WhatsApp, TikTok and a call each remove different evidence.

A method that needs detail the road removed cannot work, however accurate it was in a paper. We choose methods by what remains.

Bandwidths: Opus RFC 6716, AMR-WB 3GPP TS 26.171 · TikTok audio: Waddell, CU Denver thesis, 2025 · C2PA metadata is not carried in Ogg Opus (C2PA specification)

Sound waves

When you speak, you push air molecules together and apart hundreds of times a second. A microphone measures that pressure; the computer stores it as a long list of numbers.



The whole clip. Seven seconds of clip A. Height is pressure: tall parts are loud, flat parts are pauses.

Zoom to 40 ms. One vowel. The same shape repeats every 3.5 ms; each repeat is one opening and closing of the vocal folds.

Zoom to 2 ms. The smooth curve is really dots: 32 measurements, each a whole number between −32,768 and 32,767.

From the repeat to the pitch

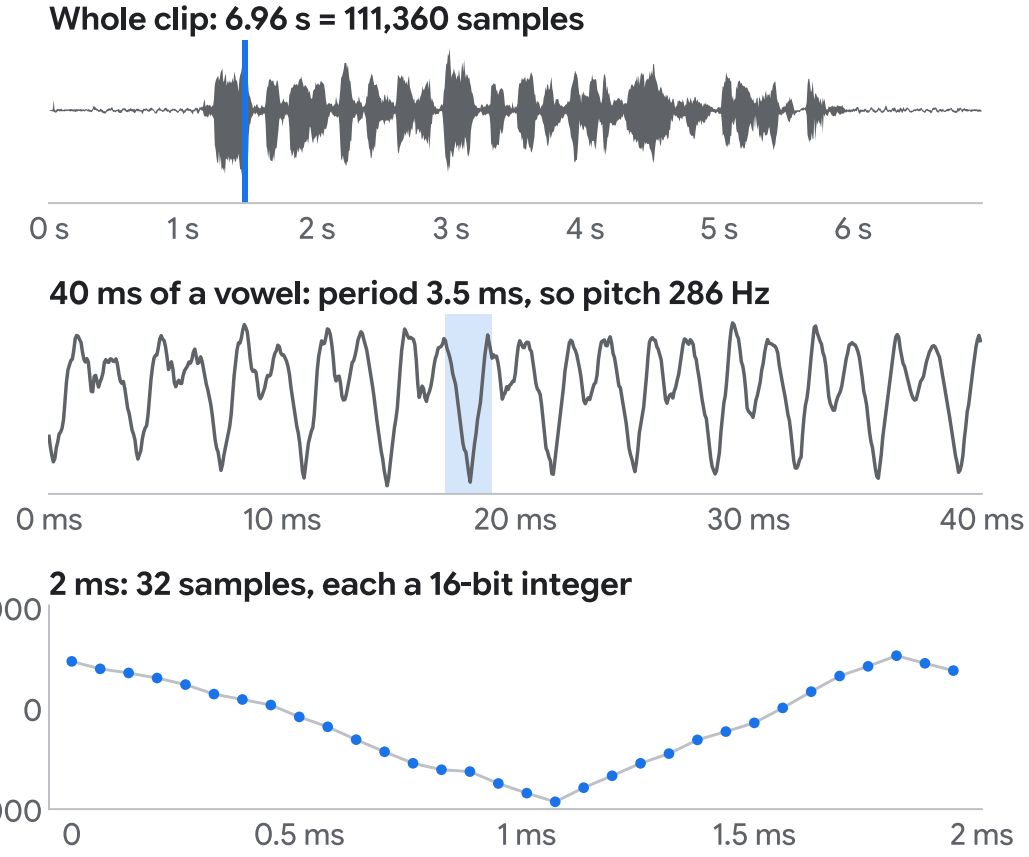
period	T = 3.5 ms = 0.0035 s for one repeat
frequency	$f = 1/T$ repeats per second, in hertz (Hz)
her pitch	**1 / 0.0035 s = 286 Hz, in this vowel**

Your turn. A repeat every 7 ms: what pitch?

Clip A: a speaker from Google FLEURS (CC BY 4.0), decoded at 16 kHz · the same zoom is cell 2 of Stop 1

Sound waves

When you speak, you push air molecules together and apart hundreds of times a second. A microphone measures that pressure; the computer stores it as a long list of numbers.



The whole clip. Seven seconds of clip A. Height is pressure: tall parts are loud, flat parts are pauses.

Zoom to 40 ms. One vowel. The same shape repeats every 3.5 ms; each repeat is one opening and closing of the vocal folds.

Zoom to 2 ms. The smooth curve is really dots: 32 measurements, each a whole number between −32,768 and 32,767.

From the repeat to the pitch

period	T = 3.5 ms = 0.0035 s for one repeat
frequency	$f = 1/T$ repeats per second, in hertz (Hz)
her pitch	**1 / 0.0035 s = 286 Hz, in this vowel**

Your turn. A repeat every 7 ms: what pitch? **1 / 0.007 s = 143 Hz.**

Clip A: a speaker from Google FLEURS (CC BY 4.0), decoded at 16 kHz · the same zoom is cell 2 of Stop 1

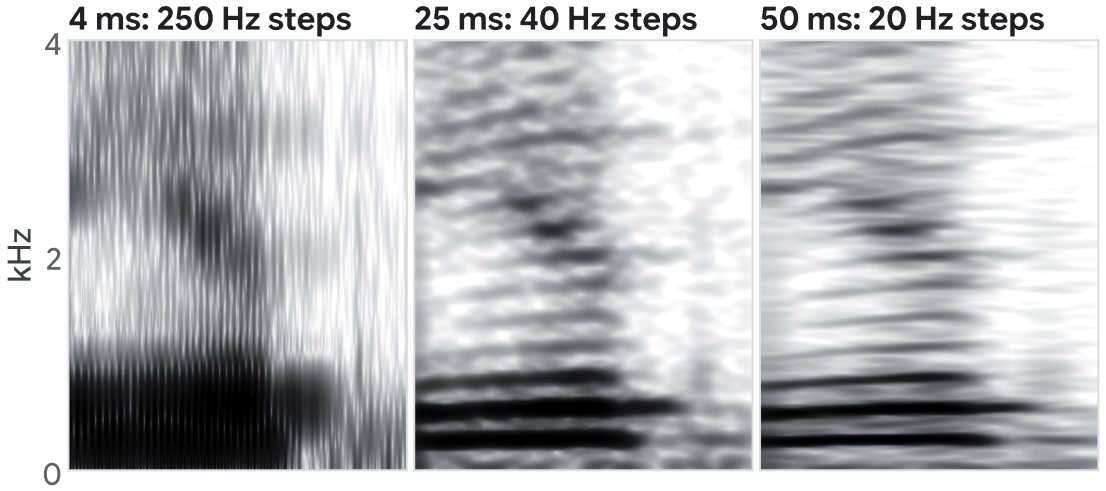
The spectrogram

A spectrogram cuts the sound into short overlapping windows, finds the recipe of each one, and lays the recipes side by side. It is a picture of which frequencies are present at each moment.

Four steps

1. Cut 25 ms windows, a new one every 10 ms. Seven seconds give about 700 windows.
2. Fourier-transform each window: 201 energies from 0 to 8 kHz.
3. Convert energy to decibels, $L = 10 \log_{10}(P/P_0)$, where P_0 is a reference power. Every 10 dB is ten times the power.
4. Draw each window as one column: time runs left to right, frequency upwards, darker means more energy.

Why decibels? Speech spans a millionfold range of power. A log scale fits quiet and loud in one picture, as the ear does.



The window is a trade-off. Short windows (4 ms) show every pulse of the vocal folds as a vertical line but blur frequency. Long windows (50 ms) separate every harmonic but blur time. Speech analysis settles on about 25 ms.

Your turn. To see exactly when a “t” bursts, would you choose a short or a long window?

Short-time Fourier transform with Hann windows · the same second of clip A in all three panels · wide-band and narrow-band spectrograms: Ladefoged & Johnson, A Course in Phonetics

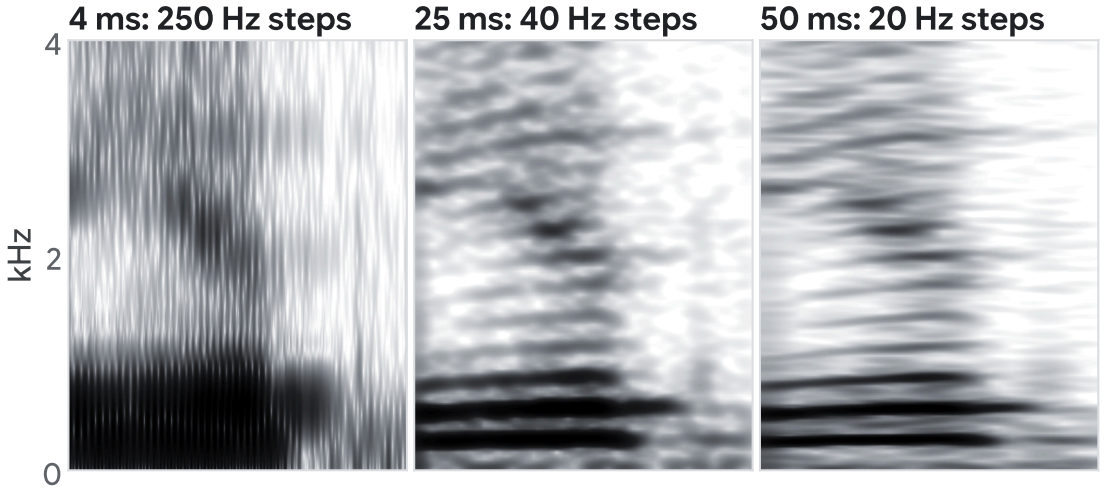
The spectrogram

A spectrogram cuts the sound into short overlapping windows, finds the recipe of each one, and lays the recipes side by side. It is a picture of which frequencies are present at each moment.

Four steps

1. Cut 25 ms windows, a new one every 10 ms. Seven seconds give about 700 windows.
2. Fourier-transform each window: 201 energies from 0 to 8 kHz.
3. Convert energy to decibels, $L = 10 \log_{10}(P/P_0)$, where P_0 is a reference power. Every 10 dB is ten times the power.
4. Draw each window as one column: time runs left to right, frequency upwards, darker means more energy.

Why decibels? Speech spans a millionfold range of power. A log scale fits quiet and loud in one picture, as the ear does.



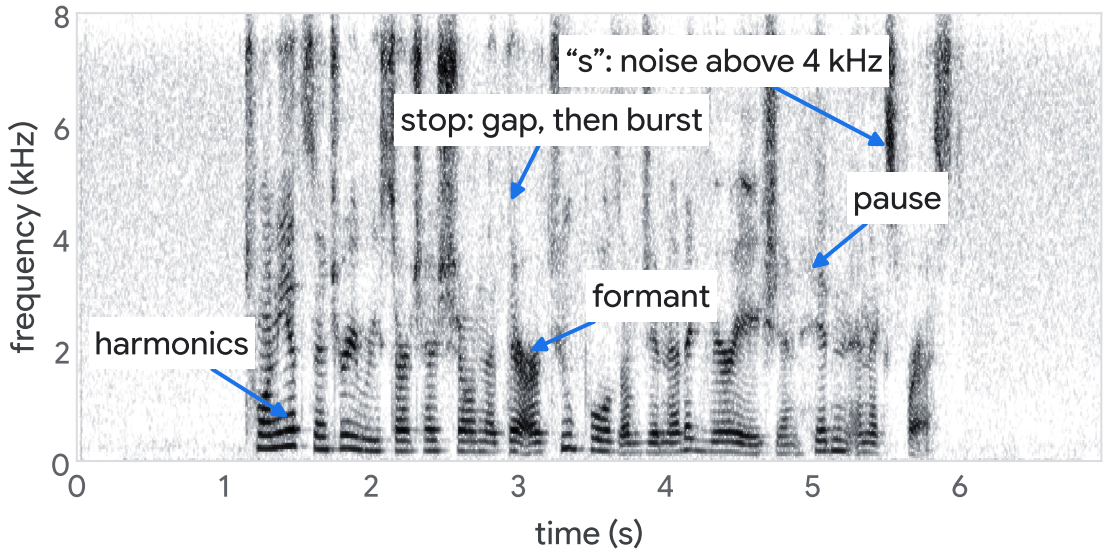
The window is a trade-off. Short windows (4 ms) show every pulse of the vocal folds as a vertical line but blur frequency. Long windows (50 ms) separate every harmonic but blur time. Speech analysis settles on about 25 ms.

Your turn. To see exactly when a “t” bursts, would you choose a short or a long window? **Short: the burst lasts a few ms.**

Short-time Fourier transform with Hann windows · the same second of clip A in all three panels · wide-band and narrow-band spectrograms: Ladefoged & Johnson, A Course in Phonetics

Reading a spectrogram

Phoneticians read spectrograms almost like text. Five patterns cover most of what you will see. Here they are in clip A: "… research has shown that there are two pools of genetic variation: hidden and expressed."



Harmonics or formants? Harmonics are the thin stripes, whole multiples of the pitch. Formants are the broad dark regions where the throat and mouth make some harmonics louder. A higher pitch spreads the stripes; moving the tongue moves the formants.

1. **Voicing:** evenly spaced stripes, the harmonics, spaced by the pitch (F0). The vocal folds are vibrating.
2. **Vowels:** dark horizontal bands, the formants. They slide up and down as the tongue moves.
3. **Hissing sounds** such as s, sh and f: grey noise without stripes. The s is darkest above 4 kHz.
4. **Stops** such as p, t and k: a short blank gap while the mouth is closed, then a thin vertical burst.
5. **Pauses:** pale columns from bottom to top.

Reading procedure after R. Hagiwara, "How to read a spectrogram", University of Manitoba · formants versus harmonics: B. Story, ASHA, 2024 · clip A, 25 ms windows, 55 dB range

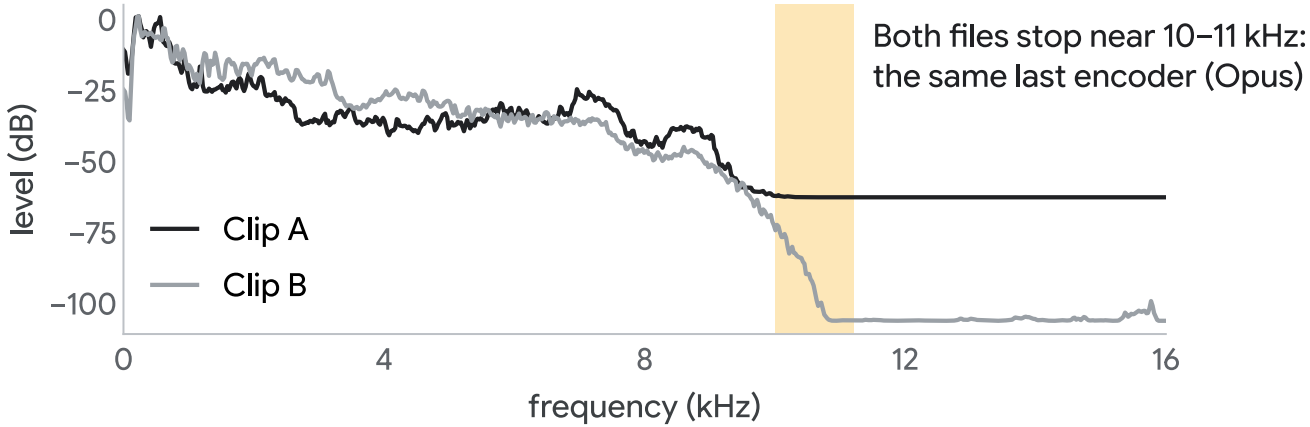
Colab Stop 1

Listen to both clips, then measure them: their spectrograms on one colour scale, their average spectrum, and the file information that ffprobe, a free tool, reads from the file's wrapper (its container).

File information, read by ffprobe

```
A.ogg · 12,432 bytes · 6.97 s
  container Ogg
  stream 0 Opus, mono
    decoder clock 48,000 Hz
    input-rate field 16,000 Hz
    encoder Lavc60.31.102 libopus
B.ogg · 18,184 bytes · 6.65 s
  same container, codec, encoder
```

Bitrate $= 8 \times$ bytes/seconds: A is $8 \times 12\,432/6.97 = 14.3$ kbit/s, B is 21.9 kbit/s. One encoder wrote both files, so the metadata names the last program that touched them, not the source of the voice.



A was recorded at 16 kHz, so its energy above 8 kHz was painted in by the Opus encoder. Real WhatsApp notes stop near 8 kHz.

In the notebook

1. Play A and B; compare both spectrograms on the same scale.
2. Read the file information and the average spectrum.
3. **Chat:** one difference you can measure, not only hear.

Opus decoders always run at 48 kHz; the input-rate field is informational only (RFC 7845) · both clips were re-encoded for the workshop with ffmpeg and libopus

Anatomy of the voice

Speech is air from the lungs, set vibrating by the vocal folds in the larynx, then shaped by a tube about 17 cm long that the tongue, jaw, soft palate (velum) and lips reshape many times a second.

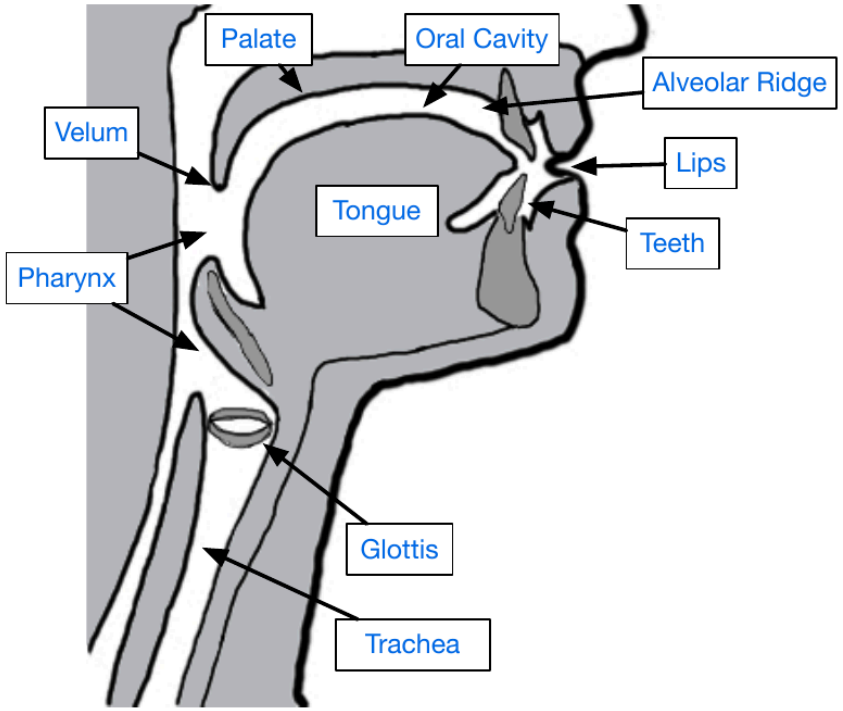


Fig. 1 from Blue et al., USENIX Security 2022: the parts of the vocal tract. The vocal folds sit at the glottis, behind the Adam's apple.

Power. The lungs push air at a pressure of roughly 5 to 10 cm of water during normal speech. Breath sets how long a phrase can last.

Sound source. In the larynx, behind the Adam's apple, two vocal folds open and close about 80 to 300 times a second. Each puff of air is one cycle of the voice.

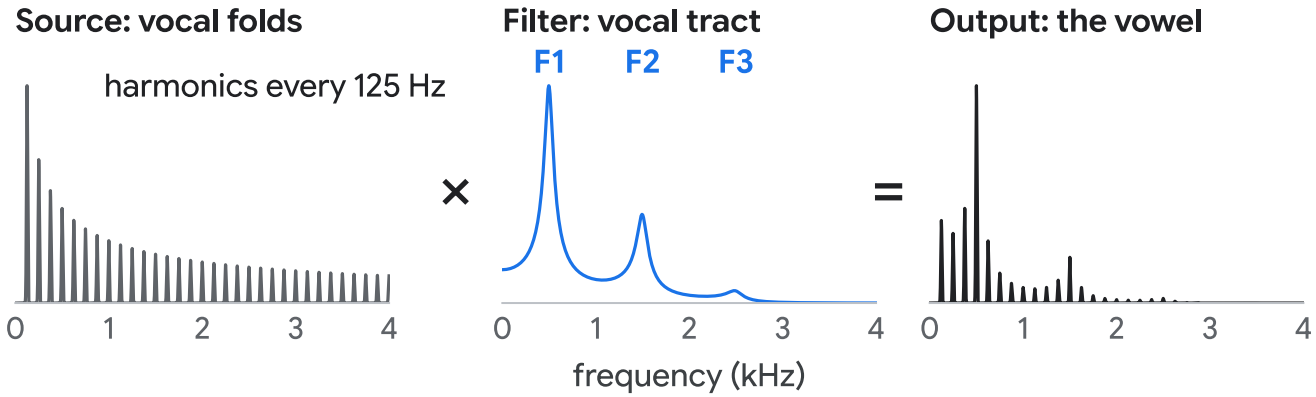
Filter. Above the larynx, the throat (pharynx), mouth and nose act as resonating tubes. Moving the tongue changes their shape and so the vowel.

Try it now. Put two fingers on your Adam's apple. Say "zzzz", then "ssss". You feel the folds vibrate only for z: same mouth, different source.

Titze, Principles of Voice Production, 1994 · Fant, Acoustic Theory of Speech Production, 1960 · figure: Blue et al., "Who Are You (I Really Wanna Know)?", USENIX Security 2022, reproduced for teaching

Source and filter

Voiced speech is two things multiplied: a buzz from the vocal folds (the source) and the resonances of the tube above them (the filter). Pitch comes from the source; the vowel comes from the filter.



Source. The folds open and close F_0 times a second, so the buzz contains every multiple of F_0: the harmonics. Here F_0 = 125 Hz, so harmonics at 125, 250, 375 Hz and up.

Filter. The tube boosts harmonics near its resonances and weakens the rest. Output = source × filter, frequency by frequency.

Not the same thing. Harmonics are set by the folds; formants by the tube's shape. A formant is a region, not a single stripe.

Try it. Whisper "aah". A whisper has no buzz at all, only rushing air, yet you still hear "aah". The filter alone carries the vowel.

Try it. Sing "aah" low, then high. The pitch changes; the vowel stays "aah".

Why a detector cares. Speech generators model pitch and spectrum in separate parts of the network. In a body they come from one throat and stay consistent; in a model they can drift apart.

In Colab. The extension cell after Stop 1 builds "aa", "ee" and "oo" from one buzz and three filters, and plays them.

Source–filter theory: Fant, Acoustic Theory of Speech Production, 1960 · explaining formants versus harmonics: Story, ASHA, 2024 · F0 125 Hz and a 17 cm tube in the chart

What a generator must imitate

Every trace in the first column had to come from a body. A generator has to imitate each one, and each one becomes numbers that our detector measures.

Trace	Made by	How we measure it	Kept by WhatsApp?	In our detector
Pitch melody	larynx, planning	pitch track: range, spread, slope, voiced share	yes	4 voicing numbers
Cycle irregularity	vocal folds	jitter, shimmer, harmonics-to-noise ratio	partly: codecs add their own	3 voicing numbers
Formants and their motion	tongue, jaw, lips	cepstral coefficients (Detector section)	yes, below 8 kHz	106 cepstral numbers
Breaths	lungs	quiet, low-frequency gaps of 150–800 ms	often hidden by noise	2 breath numbers
Pauses and rhythm	planning, breathing	silent runs, loudness rhythm	yes	5 timing numbers
Fine detail above 4 kHz	mouth, room, microphone	energy and its shape in high bands	no on calls; partly on WhatsApp	15 spectral numbers
Words and dialect	the person	listening, asking, calling back	yes	not in any detector

How text becomes speech

A voice-cloning text-to-speech system is three networks in a row. One learns who is speaking, one decides what the speech should look like, and one turns that picture into sound.

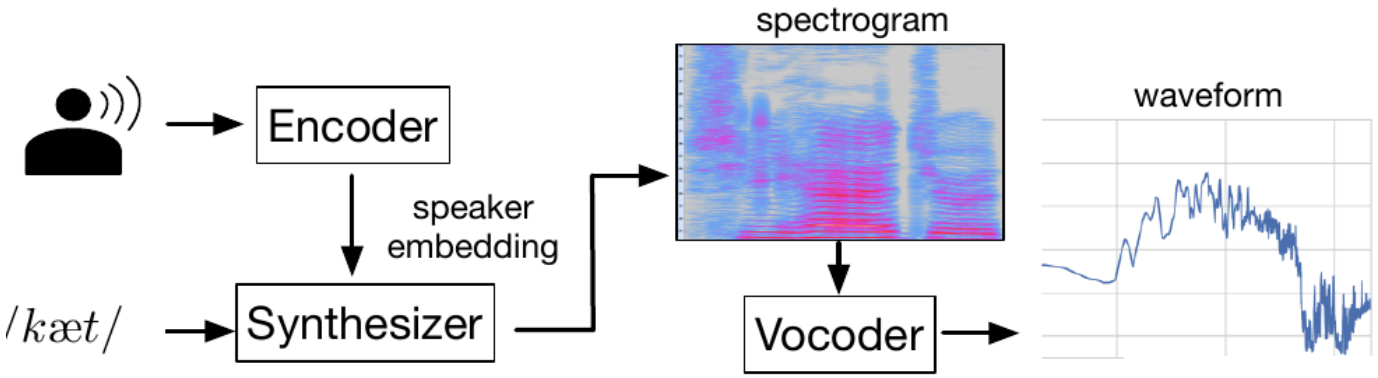


Fig. 2 from Blue et al., USENIX Security 2022.

Voice conversion skips the text: the scammer speaks and a model swaps his timbre for the target's. His breathing and melody stay human, so detection must rely on the voice source and the spectrum.

1 Speaker encoder. A network trained to tell voices apart turns a few seconds of the target into an *embedding*: a few hundred numbers that summarise the voice.

2 Synthesizer. Text is written out ("AED 500" becomes "five hundred dirhams"), turned into phonemes, the sound units of a language (/kæt/ is "cat"), and drawn as a spectrogram in that voice.

3 Vocoder. The spectrogram keeps energy per band every 12 ms, not the wave. The vocoder invents what is missing: every cycle, every phase.

Blue et al., "Who Are You (I Really Wanna Know)?", USENIX Security 2022, Fig. 2 reproduced for teaching · listeners rated Tacotron 2 at 4.53 out of 5 against 4.58 for real speech (Shen et al., ICASSP 2018)

Features

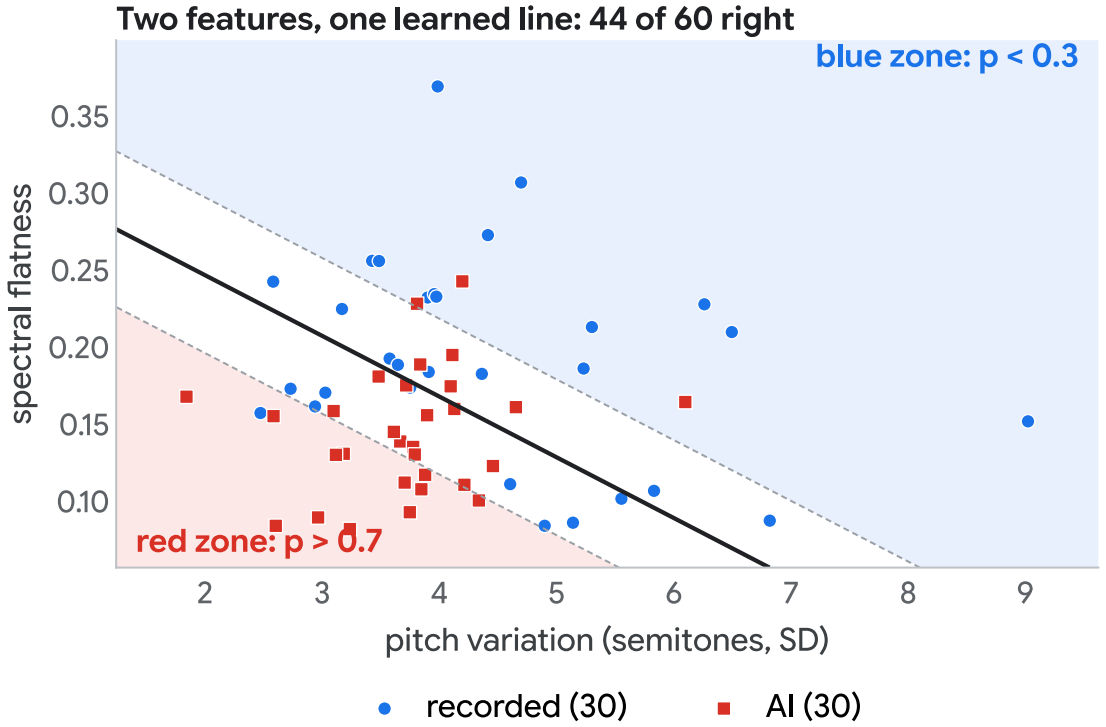
A detector cannot listen. We describe every clip with the same list of measurements, its feature vector, and let a model learn which values go with which label.

whole-clip feature	clip A, recorded	clip B, AI
jitter	2.53%	1.70%
shimmer	13.5%	7.5%
harmonics-to-noise ratio	10.7 dB	14.6 dB
pitch range	9.4 semitones	13.6 semitones

4 of the 135 numbers in detector v2, measured over the whole clip, so jitter is higher than over 9 steady cycles.

Feature vector. One row of numbers per clip, $\mathbf{x} = (x_1, \dots, x_d)$.
Label. $y = 0$ for recorded, $y = 1$ for AI. Our first detector uses $d = 58$: a table of 60 rows and 58 columns.

Not every number agrees. B varies its pitch more than A, against the trend. The model weighs them all.

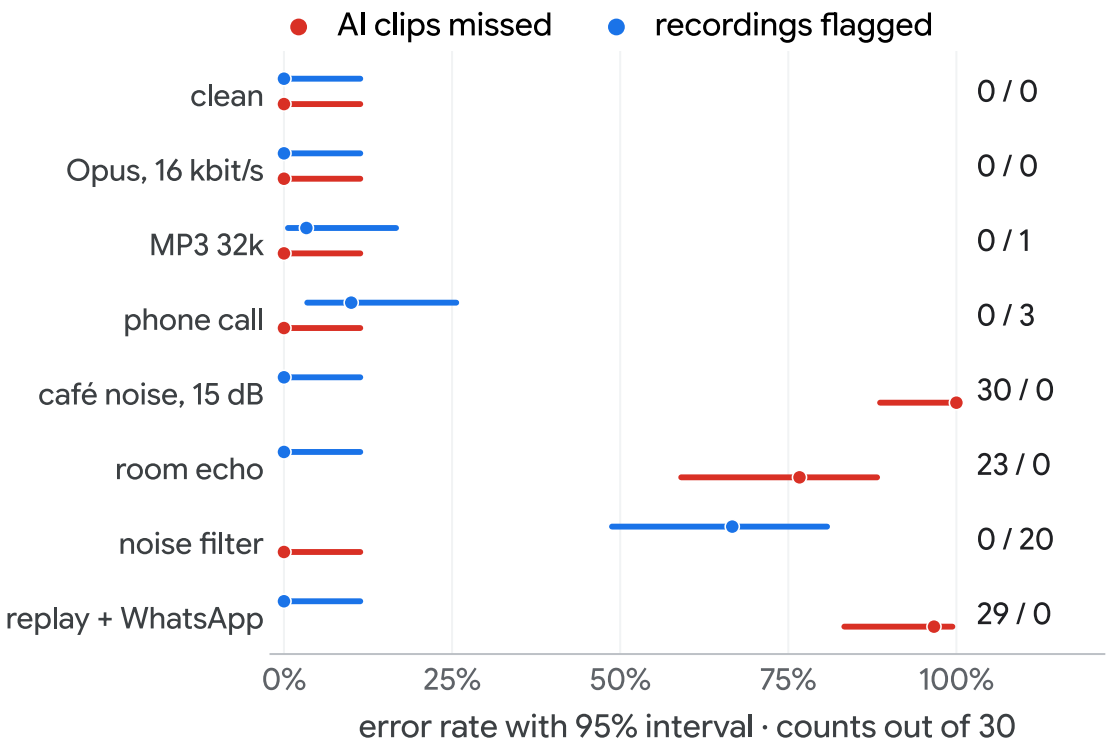


Each dot is a clip, placed by two of its numbers. Between the dashed lines: inconclusive.

Feature values from the workshop notebook (Praat via Parselmouth) · two-feature model fitted on the 60 clips with scikit-learn

Colab Stop 2

Train the first detector, 58 numbers and logistic regression, with five grouped folds. Then test it on eight roads it never saw in training. A perfect clean score is where the questions start.



Dot: measured error rate. Bar: 95% Wilson interval. With 30 clips, 4 errors fit any true rate from 5% to 30%.

Clean clips: 60 of 60

	called recorded	called AI
recorded	30	0
AI	0	30

Too good to trust. One dataset of recordings, three AI programs: the model may be recognising the source.

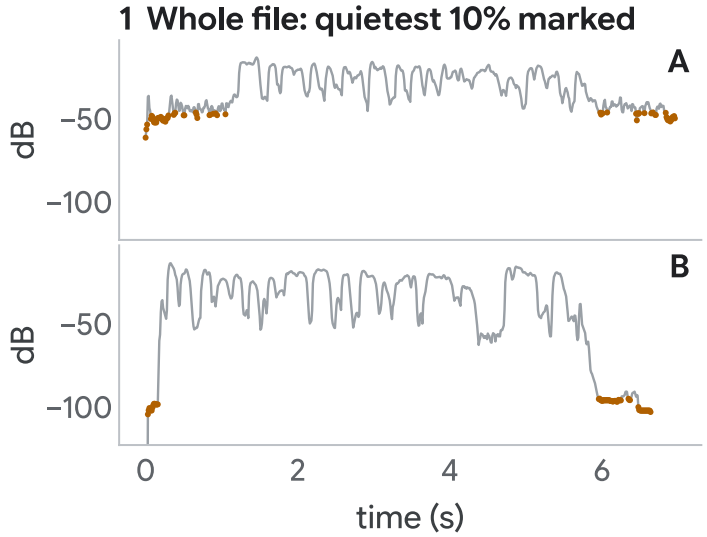
$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + z^2/n}, \qquad \hat{p} = \frac{k}{n},\ z = 1.96$$

k errors out of n = 30 clips. Café noise at 15 dB: speech about 32 times stronger than the noise.

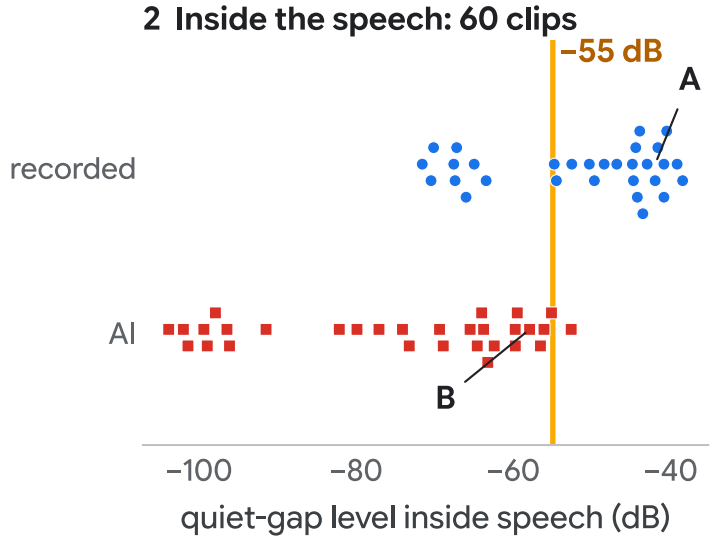
Chat: which road worries you most?

Shortcut learning

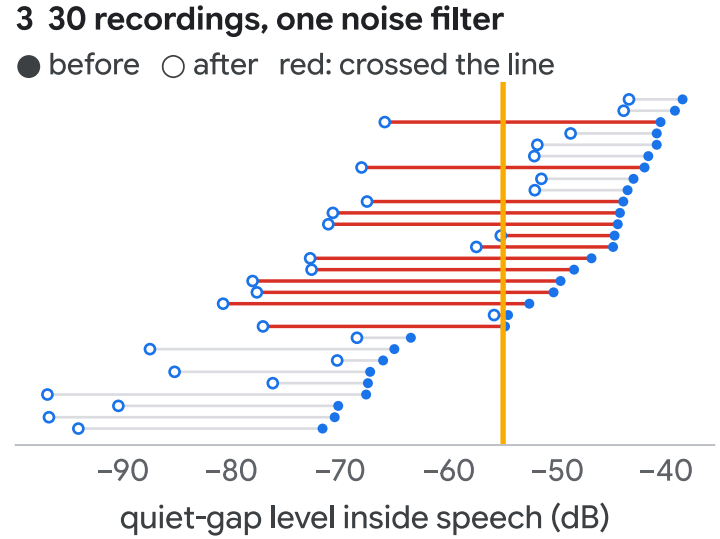
The quiet-gap clue: AI clips seem to have cleaner silence. Measured three ways, it turns out to describe how each file was recorded and prepared, not who or what was speaking.



Across the whole file, the quietest frames are the file edges. B's edges are digital silence: A −51 dB, B −105 dB.



Inside the speech, a line at −55 dB catches 29 of 30 AI clips but also flags 9 of 30 recordings.



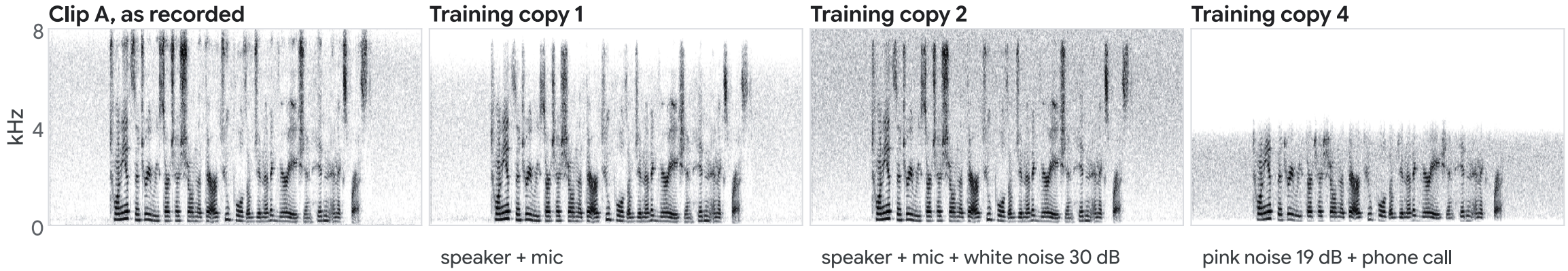
One noise filter, the kind phones and Teams apply, raises the flagged recordings from 9 to 23 of 30.

The same problem exists in research: on the ASVspoof 2019 benchmark, the length of the silence at the start of a clip alone classified clips with about 85% accuracy.

Measured on the 60 workshop clips; noise filter: FFmpeg afftdn · silence shortcut in ASVspoof 2019: Müller et al., "Speech is Silver, Silence is Golden", ASVspoof Workshop 2021

Training on the road

If every recording in training has room noise and no AI clip does, noise becomes a label. The fix is to give both classes the same random roads, so the road carries no information about the answer.



Copy 3 of clip A drew no change at all: randomness includes the clean case.

Random roads. For each copy we roll dice: speaker and microphone (40%), room echo (40%), noise 8 to 30 dB below the speech (50%), a noise filter (20%), then Opus, MP3, a phone call or nothing.

Both classes, same dice. Recordings and AI clips get identical treatment. Noise, echo and codecs now appear equally on both sides, so the model can no longer use them as a shortcut.

The arithmetic. 60 clips × (1 original + 6 copies) = 420 training examples. Test clips get their own eight fixed roads and are never used for training.

Data augmentation for anti-spoofing: Tak et al., “RawBoost”, ICASSP 2022 · the roads are the random_channel function in the notebook; these are the copies it made of clip A

Our detector, version 2

We build a detector that a Colab CPU can build in a few minutes, that we can explain line by line, and that is trained on the roads it will meet, not only on clean files.

1 Decode

Any format to **16 kHz mono**.

Trim silence at both ends, normalise the peak level.

→ 2 Augment

Training only. Six random roads per clip: codec, noise, room, phone band, noise filter.

Applied to recordings and AI clips alike.

→ 3 Measure

135 numbers per clip: cepstra, spectral shape, voicing, pauses and breaths, rhythm.

→ 4 Weigh

Standardise each number, then **logistic regression** with strong regularisation (C = 0.05).

→ 5 Decide

$p < 0.3$ consistent with a recording

0.3 to 0.7 inconclusive

$p > 0.7$ consistent with AI

Feature groups. Cepstra (MFCC, and LFCC: equally spaced bands) describe the envelope; spectral shape catches vocoder energy; voicing measures the folds; pauses and rhythm measure timing. 135 numbers per clip.

Augmentation. During training every clip is copied six times through random codecs, noise, rooms, phone bands and noise filters. Recordings and AI clips get the same treatment, so the road stops carrying the label.

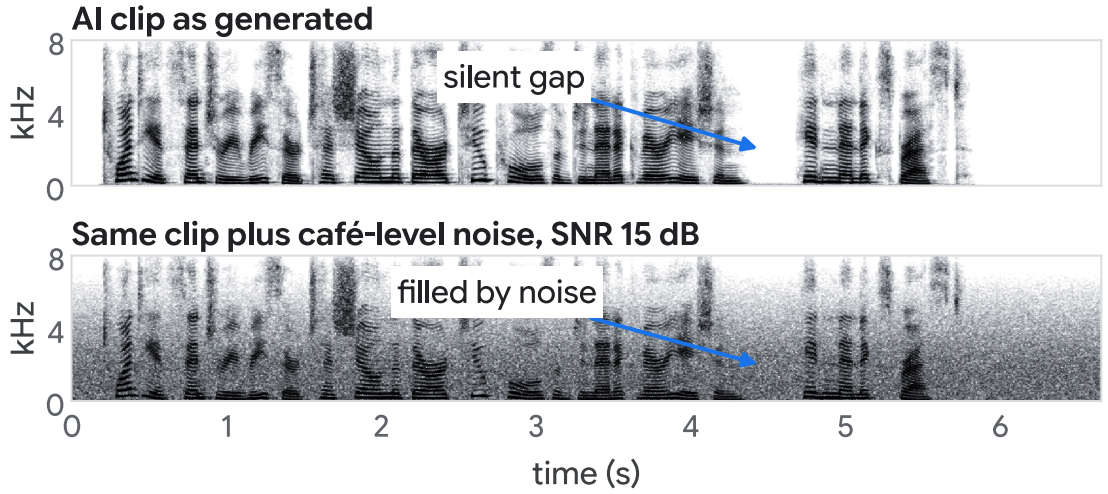
A simple, honest model. Logistic regression with strong regularisation: 135 weights we can inspect. Three zones instead of yes or no.

MFCC 26 · LFCC 40 · LFCC change from frame to frame 40 · spectral shape 15 · pauses and breaths 5 · rhythm 2 · voicing 7 · about 3 minutes on a Colab CPU

LFCC: Sahidullah, Kinnunen & Hanilçi, Interspeech 2015 · augmentation of both classes: Tak et al., RawBoost, ICASSP 2022 · voicing features: Praat via Parselmouth · code in the workshop notebook

Colab Stop 3

Test the first detector with café noise, then build detector v2: 135 measurements, trained on six random roads per clip, tested on the same eight roads as Stop 2.



Café noise. The same AI clip before and after café noise: the first detector's score falls from 0.91 to 0.04. Every training recording had noise in its pauses, so it had learned "noise means human".

$\mathrm{SNR} = 10 \log_{10}(P_s/P_n) = 15$ dB: the speech has about 32 times the power of the noise.

In the notebook

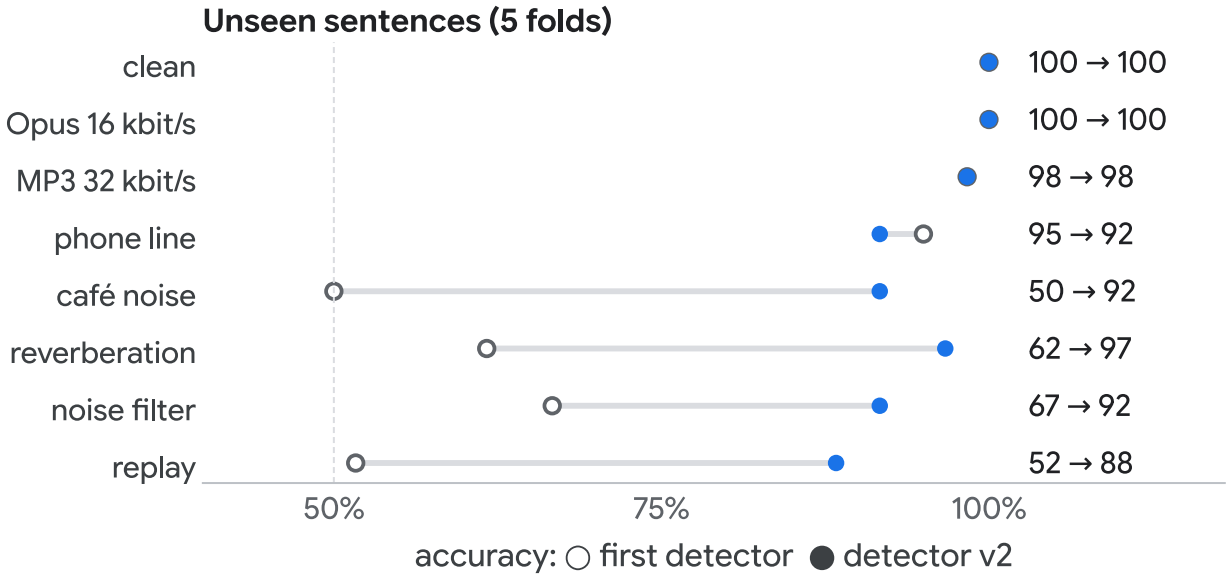
1. Run the noise cell. Predict the score before you look.
2. Read the three feature cells: spectral, voicing, timing.
3. Measure 60 clips × 14 versions: about 3 minutes on a CPU.
4. Train v2 and compare both detectors on all eight roads, in both tests.
5. **Chat:** which road still hurts v2 the most, and why?

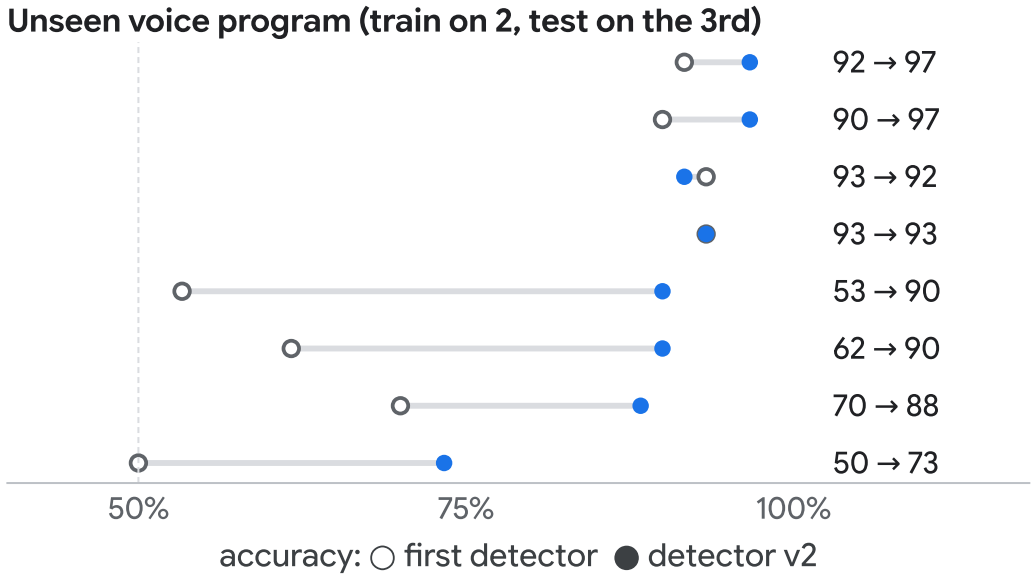
Fast finishers. Change `N_ROADS` to 2, or `C` to 1.0 (weaker regularisation), rerun, and see which roads get worse.

Results from the workshop notebook: clip B scored by a first detector that never saw its sentence · noise: speech-band babble at 15 dB SNR

Detector v2 results

We tested both detectors on clips they never saw, under eight conditions, in two ways: new sentences, and a whole voice program held out of training.





Robust to the road. Under café noise the first detector guessed (50%); v2 is right 92% of the time. Under simulated replay, 52% becomes 88%.

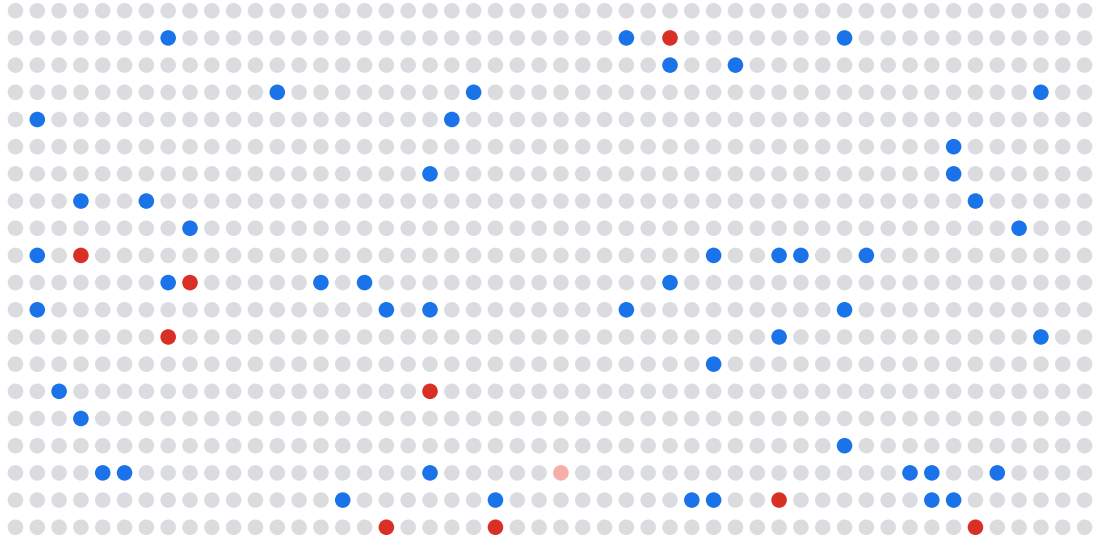
New generators are harder. With one voice program never heard in training (and 10 recorded sentences held out with it), v2 keeps 88–97% on most roads, 73% after replay.

What we can claim. 60 clips, three generators, simulated roads. That is a teaching result, not a product: a later slide puts it next to published systems.

Accuracy at threshold 0.5 on all 60 clips, each scored by a model that never saw its sentence (left) or its voice program (right) · detector v2: 135 features, augmentation ×6, logistic regression (C = 0.05)

Base rates

A detector's accuracy means little until you know how common fakes are where it is used. When fakes are rare, most of its flags are false alarms.

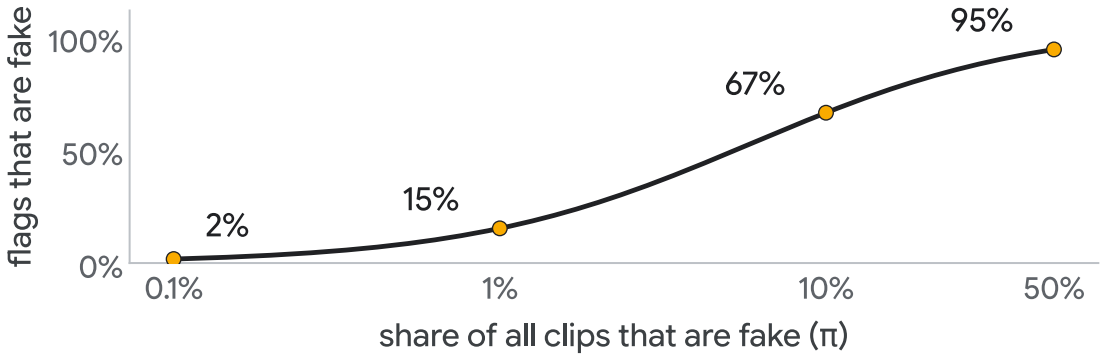


● 9 fakes caught ● 1 missed ● 50 real flagged ● 940 real passed

1,000 voice notes, 1% fake, a detector that catches 90% of fakes and flags 5% of real notes: 59 flags, 9 of them fake.

$$P(\text{fake} \mid \text{flag}) = \frac{\text{TPR}\,\pi}{\text{TPR}\,\pi + \text{FPR}\,(1-\pi)}$$

$$= \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.05 \times 0.99} = \frac{0.009}{0.0585} \approx 15\%$$

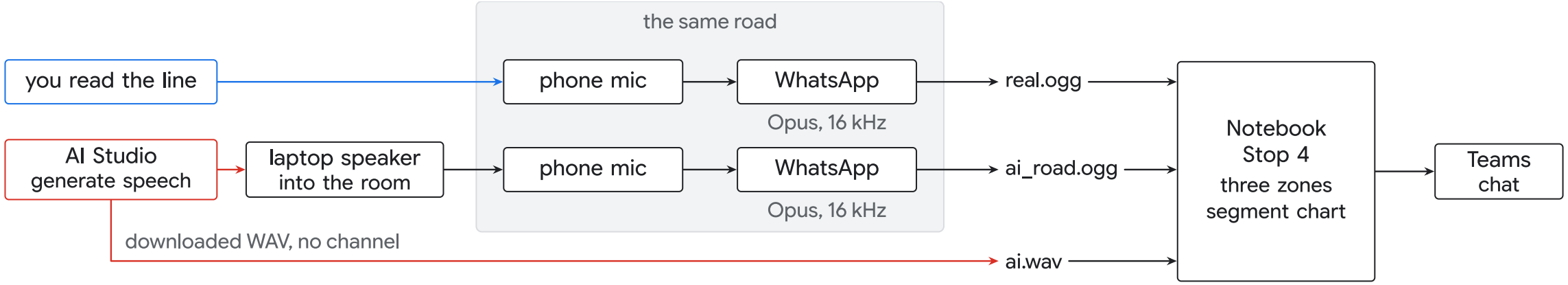


π: share of clips that are fake. TPR: share of fakes caught (1 − miss rate). FPR: share of real clips flagged (the false-alarm rate). The same detector is right 95% of the times it flags when half the clips are fake, 2% when one in a thousand is.

Bayes' theorem with illustrative rates (TPR 90%, FPR 5%) · NIST AI 100-4 (2024), Reducing Risks Posed by Synthetic Content

Colab Stop 4

Make one real and one AI voice note that travel the same road, then run both through the detector. The hard case is the realistic one: same phone, same app, same room.



“Hi, it’s me. I’m on a friend’s phone. Can you send me the code you just got?”

Getting the files into Colab: send both notes to yourself, open web.whatsapp.com on the laptop, download them, then upload them in the last notebook cell.

Rules. Use only your own voice or AI Studio’s stock voices. Colab’s terms do not allow creating deepfakes inside Colab, so generation happens in AI Studio.

Report. The notebook prints a report for each file: zone, windows, voicing and breathing against the 60 clips. Post the zone and whether the windows agree.

Compare. Upload the same files to Deepfake Total (Fraunhofer AISEC), a research detector that accepts up to 90 s, and set its answer beside ours.

Your report

The answer to the CEO voice note is not “fake” or “real”. It is a short report that someone else can check, with your limits written down.

Material	Opus file, 9.4 s, forwarded twice; SHA-256 hash noted; no original
Road	Opus 16 kHz: nothing above 8 kHz; WhatsApp metadata only
Checks	listening, detector v2 and segments, breathing, dialect, call back
Results	score 0.83, segments mostly high; no breath heard (weak in 9 s); unusual phrasing; CEO denies it on call back
Limits	short clip; detector trained on 60 English clips; replay not excluded
Conclusion	inconsistent with an original recording of the CEO; do not transfer
Next steps	keep the file and its hash; report to the fraud unit and police e-crime

Why this format. Forensic standards ask for the material, the methods, the results and the limits, so another examiner can repeat the work.

The strongest line is the call back, not the score. The detector supports the conclusion; it does not make it.

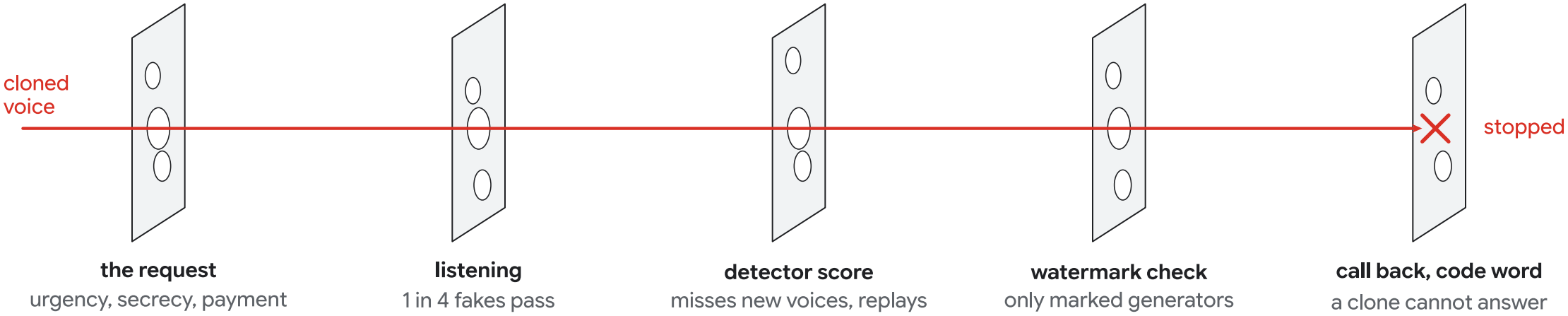
The graded scale, consistent, inconclusive or inconsistent, is honest about uncertainty in a way “99% fake” is not.

The hash is a fingerprint of the file: change one bit and it changes. Writing it down proves later that the file you examined is the file you received.

Report structure after SWGDE Best Practices for Digital Audio Authentication and ENFSI FSA-BPM-002 (2022) · example values are illustrative

Verification

Every layer has holes, and a scam succeeds when the holes line up. The one check a cloned voice cannot pass is a call back on a number you already have, or a code word agreed in person.



Holes line up. Listening, a detector and a watermark check each fail sometimes. A scam succeeds when all of them fail on the same clip.

The check a clone cannot pass. A call back on a number you already have, or a code word agreed in person, does not depend on the audio at all.

Tonight. Agree a code word with your family, in person. Before you approve any transfer, call back on a number you already have.

Reason, Human Error, 1990 · FBI PSA I-120324 (Dec 2024): hang up, call back, agree a family code word · Dubai Electronic Security Center (Sep 2026): call or visit before any transfer

Thank you

Thank you for spending the evening with GDG On Campus Abu Dhabi University. Scan the code to join our WhatsApp community for the next workshops.

Tonight: agree a code word with your family, in person.

Keep exploring: the slides and the Colab notebook stay pinned in the Teams chat; Run all works any time.

Questions: ask now, or message us in the community.

GDG On Campus ADU · WhatsApp community